Coupling between an Optical Phonon and the Kondo Effect

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We explore the ultrafast optical response of $Yb_{14}MnSb_{11}$, providing further evidence that this compound is the first *d*-electron, ferromagnetic, underscreened Kondo lattice. These results also provide the first demonstration of coupling between an optical phonon mode and the Kondo effect.

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For over 40 years, there has been significant interest in the screening of local moments by conduction electrons, termed the Kondo effect. Since its discovery [1], the Kondo effect has produced a number of surprising results, including heavy Fermion (HF) compounds where the carriers are extremely massive [2]. Kondo later suggested that the conduction electrons could also screen the motion of atoms [3]. It has since been proposed that coupling between phonon modes and the Kondo effect could manifest new material properties, such as non–Fermi liquid behavior and unconventional superconductivity [4–7]. Furthermore, strong coupling between magnetism and phonon modes has produced new effects in other systems, such as colossal magnetoresistance and multiferroic behavior in transition metal oxides [8–10]. However, there have been no observations of coupling between the Kondo effect and an optical phonon mode, in part because it has been unclear how to detect such a coupling.

The lattice plays an important role in the 14-1-11 HF compounds, where various properties can be altered through isoelectronic substitutions. The 14-1-11 compounds exhibit numerous ground states, including magnetic order, metallic or semiconducting transport [11,12], and HF behavior [13,14]. Of particular interest is $Yb_{14}MnSb_{11}$, which is metallic and exhibits ferromagnetic order of the Mn d^4 local moment below a Curie temperature (T_C) of 53 K. A study of its optical conductivity suggested that the Mn *d* shell has a fifth electron that is screened by the Kondo effect. It was argued that the coexistence of ferromagnetism and HF behavior occurs via a distortion of the $MnSb_4$ tetrahedron [14], motivating us to search for Kondo-lattice coupling in $Yb_{14}MnSb_{11}$.

We demonstrate Kondo-phonon coupling in $Yb_{14}MnSb_{11}$ by simultaneously monitoring the electronic structure and phonon dynamics via ultrafast optical experiments [15,16]. Specifically, the relaxation time and amplitude of the photoinduced response strongly increase at low temperatures (T), indicating the presence of a hybridization gap in $Yb_{14}MnSb_{11}$. Furthermore, a softening of a phonon at low T is correlated with the development of this HF state. This study therefore provides a blueprint for future examination of Kondo-phonon coupling in other HF compounds.

The $Yb_{14}MnSb_{11}$ single crystals were grown as described elsewhere [11,12]. This compound has been well characterized by x-ray diffraction, magnetic susceptibility [11], electrical resistivity, optical conductivity [14], specific heat [12], and x-ray magnetocircular dichroism [17]. The sample in this study provided mirrorlike surfaces and therefore time-domain measurements were performed on the as-grown sample. The ultrafast experiments were conducted with a cavity dumped laser producing 50 fs pulse centered at 800 nm with a 2 MHz repetition rate. Comparable results were obtained with a 250 KHz regenerative amplifier (800 nm, 100 fs) coupled to a superconducting magnet, which was used to investigate the effects of magnetic field and fluence. The pump pulse was kept at a fluence of $\approx 50 \frac{\mu J}{cm^2}$.

In the left panel of Fig. 1 we plot the photoinduced change in the reflectivity ($\frac{\Delta R(t)}{R}$) as a function of time at 310 K. After the pump pulse arrives, $\frac{\Delta R(t)}{R}$ increases indicating a rise in the temperature of the Fermi surface [14] as electron-electron interactions bring the optically excited

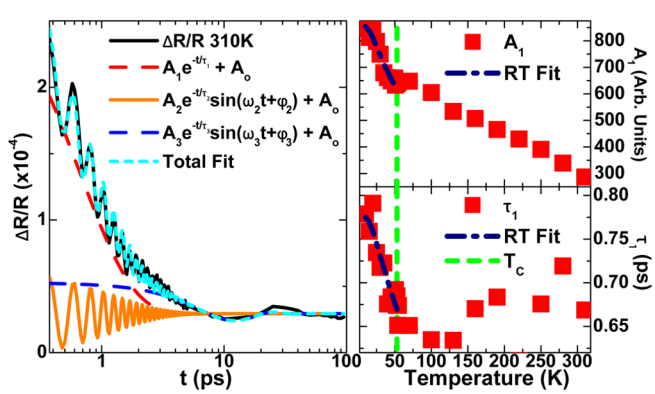

FIG. 1 (color online). Left: Pump induced change in reflectivity versus time. Data (black line) are fit with one exponential (red or gray dashed line) and two exponentially damped sinusoidal signals (orange or light gray and dashed blue or lightest gray lines). Right: Amplitude (top) and relaxation time (bottom) of the exponential term versus temperature, point size indicates error bars. A and τ increase at low T are described via a gap ($\Delta = 32\ cm^{-1}$) in the density of states for $k_B T < \Delta$ by the Rothwarf-Taylor model (navy or darkest gray dashed line).

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carriers into equilibrium [18]. We then observe an exponential decay and two exponentially damped oscillatory signals. The exponential decay is due to the carrier and phonon baths coming into thermal equilibrium [18].

To quantitatively evaluate our results we have fit the data at all temperatures to the formula:

$$\frac{\Delta R(t)}{R} = A_1 e^{-t/\tau_1} + \sum_{i=2,3} A_i e^{-t/\tau_i} \sin(\omega_i t + \phi_i) + A_0. \tag{1}$$

The resulting A_1 and τ_1 are shown in the right panels of Fig. 1. For $T > T_C$, A_1 and τ_1 follow our expectations for a normal metal. Specifically τ_1 is mostly temperature independent. However, since the specific heat of the metal is increasing as the sample temperature is raised, the photoexcitation results in a smaller change in the temperature rise of the Fermi surface. Therefore, A_1 decreases as the sample temperature is raised [18]. Interestingly, the decay time and amplitude increase significantly below T_C. Similar low temperature enhancements in τ and A (though generally larger) have been observed in superconductors [19–21] and HF compounds [15]. This low T increase is believed to indicate the opening of a gap (Δ) in the density of states. Specifically, due to the presence of a gap, the carriers can only relax via emission of high frequency phonons with energy $\hbar\omega_{\text{phonon}} \geq \Delta$. Since the probability for optical phonons to decay into two phonons of lower energy is fairly small (as opposed to acoustic modes that are much more susceptible to anharmonic affects), it is highly likely that the optical phonon will excite a new electron-hole pair across the gap. This *phonon bottleneck* significantly reduces the transmission of heat from the Fermi surface to the lattice, resulting in an increase in τ and A when $k_B T < \Delta$.

The model of Rothwarf and Taylor (RT) has been very successful in describing the dynamics of superconductors and HF compounds [22]. In the RT approach the quasiparticle and phonon dynamics are described via a set of coupled nonlinear differential equations. The RT equations account for the creation and destruction of electron-hole pairs via emission of high frequency phonons and biparticle scattering as well as a reduction in the number of high frequency phonons via diffusion and anharmonic decay. Recently the RT model has been solved in a variety of cases leading to the solution [23]

$$A(T) = \frac{A_0}{n_T(T) + 1} \tag{2}$$

$$\tau(T) = \tau_0\{\delta[\epsilon n_T(T) + 1]^{-1} + 2n_T(T)\}^{-1}, \tag{3}$$

where $n_T(T)$ is the population of the thermally excited quasiparticles. The results of simultaneously fitting $A_1(T)$ and $\tau_1(T)$ with Eqs. (2) and (3) are displayed in the right panels of Fig. 1. In performing these fits we used a standard form for the thermally excited quasiparticle density: $n_T(T) = (\Delta T)^p \exp p(\Delta/k_B T)$, where $0 < p < 1$ depends on the shape of the density of states. Similar to what has been observed in many HF compounds [15], we found $p = 0.49$ and a temperature independent gap of $\Delta = 32\ \text{cm}^{-1}$. Previous studies of the optical conductivity have found $\Delta \approx 50\ \text{cm}^{-1}$ [14]. However, optical conductivity probes the direct hybridization gap (i.e., no change in momentum), whereas the present results are governed by the indirect gap, which should be smaller [24–26]. Lastly, the application of a 6 T magnetic field, saturating the magnetization [$M(T)$], did not effect $\frac{\Delta R(t)}{R}$ at either 10 K or T_C, providing further evidence that the gap is unrelated to the ferromagnetism.

So far we have focused on the relaxation dynamics of the electronic system. However, the data displayed in Fig. 1 also contain information about an optical phonon mode. Specifically, the oscillations in the data result from the coherent generation of optical (ω_2) and acoustic (ω_3) phonons. While there are a variety of proposed mechanisms, it is generally agreed upon that such oscillations are initiated via a Raman process upon optical excitation. The modes are then observed via a Raman shift of the probe pulse, which exponentially decays due to the dephasing of the phonons [16]. As shown in Fig. 2, as the temperature is lowered, the optical mode frequency (ω_2) increases while its decay rate ($\frac{1}{\pi\tau_2}$) decreases. In contrast, for $T < T_C$ the phonon softens and broadens. As discussed later, this low temperature behavior is quite anomalous. Therefore, to confirm that this behavior is intrinsic to the sample and not an artifact of our fitting routine, we subtracted the first term of Eq. (1) from the data and then performed a fast-Fourier transform (FFT). The FFT spectra in the top panel of Fig. 3 reveal the same trends shown in Fig. 2. The center frequency and broadening determined from fits of the FFT data with Lorentzian line shapes are in superb agreement with the results of fitting $\frac{\Delta R(t)}{R}$ (see Fig. 2).

The temperature dependence of the phonon above T_C is similar to anharmonic affects observed in numerous other materials. Specifically, the optical phonon can decay into acoustic modes of lower energy, resulting in a renormlization of its self-energy. Since the likelihood of this decay depends on the occupancy of the phonon density of states, one expects a T dependence of the form

$$\Gamma_A(T) = \Gamma_0 + a\left[1 + 2n_B\left(\frac{\omega_0}{2}, T\right)\right] + b\left\{1 + 3\left[n_B\left(\frac{\omega_0}{3}, T\right) + n_B^2\left(\frac{\omega_0}{3}, T\right)\right]\right\}, \tag{4}$$

$$\omega_A(T) = \omega_0 + c\left[1 + 2n_B\left(\frac{\omega_0}{2}, T\right)\right] + d\left\{1 + 3\left[n_B\left(\frac{\omega_0}{3}, T\right) + n_B^2\left(\frac{\omega_0}{3}, T\right)\right]\right\}, \tag{5}$$

where ω_0 is the intrinsic frequency from harmonic terms,

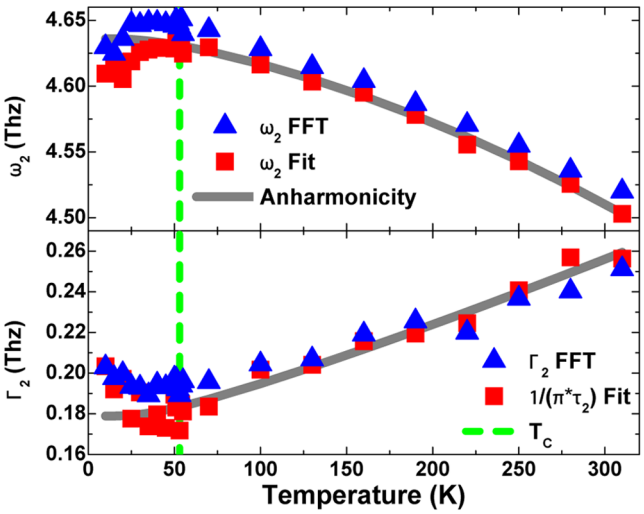

FIG. 2 (color online). The phonon mode frequency (top panel) and broadening or damping rate (bottom panel). Red or gray squares are results determined via Eq. (1), and blue or dark gray triangles are the result of fitting the FFT in Fig. 3 with a Lorentzian line shape. The mode hardens and sharpens as the temperature is lowered to T_C, which is described by anharmonic affects [Eqs. (4) and (5)].

$n_B(\omega, T) = \frac{1}{\exp(\hbar\omega/k_B T)-1}$ is the Bose occupation function, and we have assumed the phonon decays into two modes of frequency $\frac{\omega_0}{2}$ or three of frequency $\frac{\omega_0}{3}$ [27–29]. As shown in Fig. 2, Eqs. (4) and (5) describe $\omega_2(T > T_C)$ and $\Gamma_2(T > T_C)$ well. Furthermore the values of $\frac{a}{\omega_0}$, $\frac{b}{\omega_0}$, $\frac{c}{\omega_0}$, $\frac{d}{\omega_0}$ are comparable to what has been observed in Si [27–29]. Nonetheless this does not explain the T dependence of the phonon mode below T_C.

We now explore the origin of the $T < T_C$ behavior of the optical phonon, following an approach developed in other magnetically ordered systems [30–32]. It is instructive to examine the minimal mean field Hamiltonian for the underscreened Kondo lattice [33], with a phonon

$$H_{\mathrm{MF}} = \frac{\omega_0^2 x^2}{2} + \sum_{\vec{k},\sigma} \epsilon_{\vec{k}\sigma} c_\sigma c_\sigma^\dagger + J_K \sum_{\sigma\alpha i} c_{\bar{\sigma} i} d_{\bar{\sigma}\alpha i}^\dagger c_{\sigma i}^\dagger d_{\sigma\alpha i} - J_H \langle M \rangle^2, \tag{6}$$

where x is an atomic displacement, $c_{\sigma i}/d_{\sigma\alpha i}$ are the usual creation operators for c/d electrons at site i with dispersion $\epsilon_{\vec{k}\sigma}$, spin σ and orbital α. J_K is the antiferromagnetic coupling of the conduction electrons to the d electrons responsible for the Kondo effect, and J_H the ferromagnetic interaction between Mn ions with $\langle M(T) \rangle = \frac{M(T)}{M(0)}$. We can then solve for the phonon frequency since $\frac{d^2 H_{MF}}{dx^2} = \omega_2^2$, resulting in a T dependence: $\omega_2^2(T) = \omega_0^2 + \frac{d^2 J_K}{dx^2} N_{\mathrm{Kondo}}^2(T) - \frac{d^2 J_H}{dx^2} \langle M(T) \rangle^2$, where we have replaced $\sum_{\sigma\alpha i} c_{\bar{\sigma} i} d_{\bar{\sigma}\alpha i}^\dagger c_{\sigma i}^\dagger d_{\sigma\alpha i}$ by $N_{\mathrm{Kondo}}^2(T)$, the relative number of Kondo singlets [34]. Noting that we are only concerned with small atomic displacements and terms linear in x do not effect the phonon frequency, we can write $J_{H(K)}(x) \approx J_{H(K)}(0) + \beta(\gamma) x^2$. Therefore we expect the T dependence to be governed by

$$\omega_2^2 - \omega_A^2 = \beta \langle M(T) \rangle^2 + \gamma N_{\mathrm{Kondo}}^2(T). \tag{7}$$

To make use of this equation, we estimate the number of Kondo singlets from the density of quasiparticles excited across the hybridization gap: $N_{\mathrm{Kondo}}(T) = (1 - \frac{n_T(T)}{n_T(T_C)})$. We note that n_T is determined via Eq. (2) and the measured A shown in Fig. 1. This ansatz seems reasonable since the total charge is conserved as T is raised.

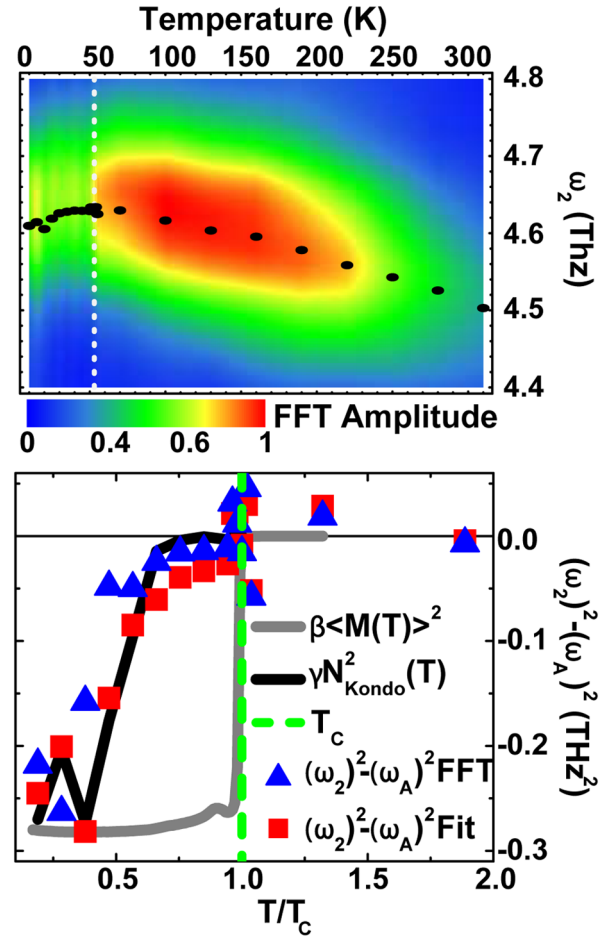

FIG. 3 (color online). Top panel: FFT of $\frac{\Delta R(t)}{R} - A_1 e^{-t/\tau_1}$, white dashed line indicates T_C and black dots are the ω_0. Bottom panel: The shift in the phonon frequency after anharmonic affects have been removed. The shift does not appear to result from coupling to ferromagnetic state (gray), but rather via a coupling to the Kondo effect (black line).

In Fig. 3 we explore the possibilities presented by Eq. (7), where we observe that the phonon continues to soften down to the lowest T. However, $\beta \langle M(T) \rangle^2$ determined via zero-field cooled SQUID measurements, saturates close to T_C, indicating the phonon does not strongly couple to the ferromagnetic order. Nonetheless, the soft-

ening of the phonon below T_C follows the temperature dependence of N_{Kondo} indicating a Kondo-phonon coupling.

Our observation of coupling of the optical phonon to the formation of Kondo singlets and not to the ferromagnetic state, may not be surprising for three reasons. First, $T_K \approx 300$ K [13,14] is much bigger than $T_C = 53$ K. Second, J_H is the exchange between the Mn atoms, which are ≈ 1 nm apart [11], whereas J_K results from the hybridization between Mn and Sb. Hence J_H should be less sensitive to small atomic displacements. Finally, the scenario proposed in Ref. [14] for the underscreened Kondo effect in $Yb_{14}MnSb_{11}$, required a distortion of the $MnSb_4$ tetrahedron in order to break the degeneracy of the d^5 electrons. Therefore one would expect a coupling between the lattice and the Kondo effect.

In conclusion, we explored the ultrafast optical properties of $Yb_{14}MnSb_{11}$. Significant changes in the dynamics occur below T_C, although they are not connected with the development of the ferromagnetic state. In particular, the amplitude and relaxation time of the electronic response grows significantly at low temperatures, but is not effected by the application of a magnetic field. Furthermore, changes in the frequency of the optical phonon are inconsistent with the temperature dependence of the magnetization. Instead we find that the low temperature enhancement of the photoinduced response and the softening of the phonon result from the development of heavy Fermions and their coupling to the lattice. As expected for a hybridization gap, quantitative analysis reveals the indirect gap probed here is smaller in magnitude than the direct gap determined via optical conductivity experiments [14]. In addition, the softening of the phonon at low temperatures is correlated with the number of heavy quasiparticles. Therefore this is the first demonstration of coupling between the Kondo effect and optical modes, and provides a guide for future studies of Kondo-Phonon coupling affects.

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